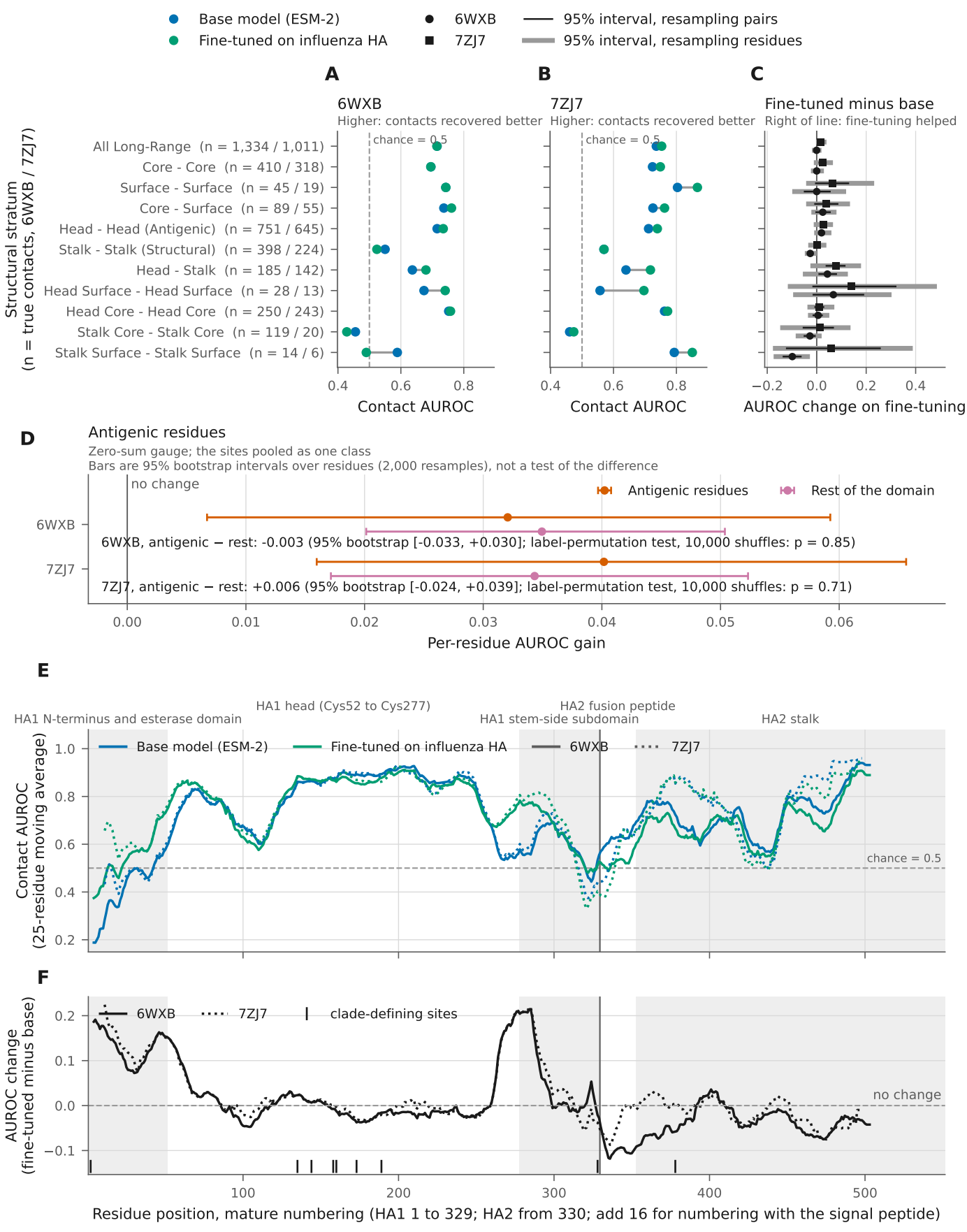

Base model (ESM-2)
Fine-tuned on influenza HA
6WXB
7ZJ7
95% interval, resampling pairs
95% interval, resampling residues
A
B
C
6WXB
Higher: contacts recovered better
7ZJ7
Higher: contacts recovered better
Fine-tuned minus base
Right of line: fine-tuning helped
Structural stratum
(n = true contacts, 6WXB / 7ZJ7)
All Long-Range (n = 1,334 / 1,011)
Core - Core (n = 410 / 318)
Surface - Surface (n = 45 / 19)
Core - Surface (n = 89 / 55)
Head - Head (Antigenic) (n = 751 / 645)
Stalk - Stalk (Structural) (n = 398 / 224)
Head - Stalk (n = 185 / 142)
Head Surface - Head Surface (n = 28 / 13)
Head Core - Head Core (n = 250 / 243)
Stalk Core - Stalk Core (n = 119 / 20)
Stalk Surface - Stalk Surface (n = 14 / 6)
chance = 0.5
Contact AUROC
AUROC change on fine-tuning
D
Antigenic residues
Zero-sum gauge; the sites pooled as one class
Bars are 95% bootstrap intervals over residues (2,000 resamples), not a test of the difference
no change
Antigenic residues
Rest of the domain
6WXB, antigenic − rest: -0.003 (95% bootstrap [-0.033, +0.030]; label-permutation test, 10,000 shuffles: p = 0.85)
7ZJ7, antigenic − rest: +0.006 (95% bootstrap [-0.024, +0.039]; label-permutation test, 10,000 shuffles: p = 0.71)
Per-residue AUROC gain
E
HA1 N-terminus and esterase domain
HA1 head (Cys52 to Cys277)
HA1 stem-side subdomain
HA2 fusion peptide
HA2 stalk
Base model (ESM-2)
Fine-tuned on influenza HA
6WXB
7ZJ7
chance = 0.5
Contact AUROC
(25-residue moving average)
F
6WXB
7ZJ7
clade-defining sites
no change
AUROC change
(fine-tuned minus base)
Residue position, mature numbering (HA1 1 to 329; HA2 from 330; add 16 for numbering with the signal peptide)